# Implicit Differentiation and Inverse Functions

ANSWER KEY

## Implicit differentiation basics

1 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.
$2x + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{x}{y}$.

2 Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.
$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$: $\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$, so $\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$.

3 Find $\frac{dy}{dx}$ for $\sin(xy) = x$.
$\cos(xy)(y + x\frac{dy}{dx}) = 1$: $y\cos(xy) + x\cos(xy)\frac{dy}{dx} = 1$, so $\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy)}$.

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## Tangent lines to implicit curves

4 Find the equation of the tangent line to $x^2 + y^2 = 25$ at the point $(3, 4)$.
$\frac{dy}{dx} = -\frac{x}{y}$; at $(3, 4)$: slope $= -\frac{3}{4}$. Tangent: $y - 4 = -\frac{3}{4}(x - 3)$, i.e. $y = -\frac{3}{4}x + \frac{25}{4}$.

5 Find $\frac{dy}{dx}$ at the point $(1, 2)$ on the curve $x^2y + y^3 = 10$.
$2xy + x^2\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$: $\frac{dy}{dx}(x^2 + 3y^2) = -2xy$, so $\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$. At $(1, 2)$: $\frac{-4}{1 + 12} = -\frac{4}{13}$.

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## Second derivatives via implicit differentiation

6 Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 1$, expressing your answer in terms of $x$ and $y$.
$\frac{dy}{dx} = -\frac{x}{y}$. Differentiate again: $\frac{d^2y}{dx^2} = -\frac{y - x\frac{dy}{dx}}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{1}{y^3}$ (using $x^2 + y^2 = 1$).

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## Derivatives of inverse functions

7 If $f(x) = x^3 + 2x + 1$, find $(f^{-1})'(4)$, using $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ and noting $f(1) = 4$.
$f'(x) = 3x^2 + 2$; $f'(1) = 5$. $(f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{5}$.

8 State the derivatives of the inverse trigonometric functions $\sin^{-1}x$, $\cos^{-1}x$, and $\tan^{-1}x$.
$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1 - x^2}}$; $\frac{d}{dx}\cos^{-1}x = -\frac{1}{\sqrt{1 - x^2}}$; $\frac{d}{dx}\tan^{-1}x = \frac{1}{1 + x^2}$.

9 Differentiate $f(x) = \tan^{-1}(3x)$.
$f'(x) = \frac{3}{1 + (3x)^2} = \frac{3}{1 + 9x^2}$.

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## Implicit differentiation with logarithmic and exponential terms

10 Find $\frac{dy}{dx}$ for $e^y = xy$.
$e^y \frac{dy}{dx} = y + x\frac{dy}{dx}$: $\frac{dy}{dx}(e^y - x) = y$, so $\frac{dy}{dx} = \frac{y}{e^y - x}$.

11 Find $\frac{dy}{dx}$ for $\ln(x + y) = x^2$.
$\frac{1+\frac{dy}{dx}}{x+y} = 2x$: $1 + \frac{dy}{dx} = 2x(x + y)$, so $\frac{dy}{dx} = 2x(x + y) - 1$.

## Related applications of implicit differentiation

12 Find all points on the curve $x^2 + xy + y^2 = 7$ where the tangent line is horizontal.
Differentiate: $2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$: $\frac{dy}{dx} = \frac{-2x-y}{x+2y}$. Horizontal tangent needs $-2x - y = 0$, i.e. $y = -2x$. Substituting into the curve: $x^2 + x(-2x) + (-2x)^2 = 7 \Rightarrow x^2 - 2x^2 + 4x^2 = 7 \Rightarrow 3x^2 = 7 \Rightarrow x = \pm\sqrt{7/3}$. Points: $\left(\pm\sqrt{7/3}, \mp 2\sqrt{7/3}\right)$.

## Finding vertical tangents implicitly

13 Find the points on the circle $x^2 + y^2 = 9$ where the tangent line is vertical (i.e. where $\frac{dx}{dy} = 0$, or equivalently where $\frac{dy}{dx}$ is undefined).
$\frac{dy}{dx} = -\frac{x}{y}$ is undefined when $y = 0$. Substituting into the circle: $x^2 = 9$, so $x = \pm 3$. Vertical tangents occur at $(3, 0)$ and $(-3, 0)$.

## Inverse function derivatives from a graph or table

14 A function $f$ is one-to-one and differentiable, with $f(3) = 10$ and $f'(3) = 4$. Find $(f^{-1})'(10)$.
$(f^{-1})'(10) = \frac{1}{f'(f^{-1}(10))} = \frac{1}{f'(3)} = \frac{1}{4}$ (since $f(3) = 10$ means $f^{-1}(10) = 3$).

15 If $g(x) = x^5 + 2x - 3$, find $(g^{-1})'(-3)$, first identifying the value $a$ such that $g(a) = -3$.
$g(0) = 0 + 0 - 3 = -3$, so $a = 0$. $g'(x) = 5x^4 + 2$; $g'(0) = 2$. $(g^{-1})'(-3) = \frac{1}{g'(0)} = \frac{1}{2}$.

## Implicit differentiation in an applied context

16 A conical tank has height and radius related by $h = 2r$ at all times as it fills, so its volume is $V = \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$. Using implicit differentiation with respect to time, find $\frac{dV}{dt}$ in terms of $r$ and $\frac{dr}{dt}$.
Differentiating both sides with respect to $t$: $\frac{dV}{dt} = \frac{2}{3}\pi(3r^2)\frac{dr}{dt} = 2\pi r^2\frac{dr}{dt}$.

## A capstone implicit differentiation problem

**17** Find $\frac{dy}{dx}$ for the curve $x^2y^2 + xy = 6$ at the point $(1, 2)$.
Differentiate: $2xy^2 + x^2(2y)\frac{dy}{dx} + y + x\frac{dy}{dx} = 0$: $\frac{dy}{dx}(2x^2y + x) = -2xy^2 - y$. At $(1, 2)$: $\frac{dy}{dx}(4 + 1) = -4 - 2 = -6$, so $\frac{dy}{dx} = -\frac{6}{5}$.

**18** Find $\frac{dy}{dx}$ for $\cos(x - y) = y\sin x$.
Differentiate: $-\sin(x - y)\left(1 - \frac{dy}{dx}\right) = \frac{dy}{dx}\sin x + y\cos x$: $-\sin(x - y) + \sin(x - y)\frac{dy}{dx} = \sin x\frac{dy}{dx} + y\cos x$.
Collecting terms: $\frac{dy}{dx}[\sin(x - y) - \sin x] = y\cos x + \sin(x - y)$, so $\frac{dy}{dx} = \frac{y\cos x + \sin(x - y)}{\sin(x - y) - \sin x}$.

**19** A satellite dish's cross-section satisfies $4x^2 + y^2 = 16$ (an ellipse). Find $\frac{dy}{dx}$ at the point $(1, 2\sqrt{3})$ on the curve.
Differentiate: $8x + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{4x}{y}$. At $(1, 2\sqrt{3})$: $\frac{dy}{dx} = -\frac{4}{2\sqrt{3}} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$.

**20** For the curve $y^2 = x^3$ (a cuspidal cubic), complete each step:

**a** Find $\frac{dy}{dx}$ implicitly. $2y\frac{dy}{dx} = 3x^2$, so $\frac{dy}{dx} = \frac{3x^2}{2y}$.

**b** Evaluate $\frac{dy}{dx}$ at the point $(4, 8)$. At $(4, 8)$: $\frac{dy}{dx} = \frac{3(16)}{2(8)} = \frac{48}{16} = 3$.

**c** Find the equation of the tangent line at that point. $y - 8 = 3(x - 4)$, i.e. $y = 3x - 4$.

**21** If $f(x) = 2x^3 + x - 5$ is one-to-one and differentiable, find $(f^{-1})'(-2)$, first identifying the input $a$ such that $f(a) = -2$.
$f(1) = 2 + 1 - 5 = -2$, so $a = 1$. $f'(x) = 6x^2 + 1$; $f'(1) = 7$. $(f^{-1})'(-2) = \frac{1}{f'(1)} = \frac{1}{7}$.

**22** For the hyperbola $xy = 6$, complete each step:

**a** Find $\frac{dy}{dx}$ using implicit differentiation. $x\frac{dy}{dx} + y = 0$, so $\frac{dy}{dx} = -\frac{y}{x}$.

**b** Confirm your answer matches what you'd get by first solving for $y = \frac{6}{x}$ and differentiating directly. Direct: $y = 6x^{-1}$, so $\frac{dy}{dx} = -6x^{-2} = -\frac{6}{x^2}$. Substituting $y = \frac{6}{x}$ into $-\frac{y}{x}$: $-\frac{6/x}{x} = -\frac{6}{x^2}$. Both match.

**c** Evaluate $\frac{dy}{dx}$ at the point $(2, 3)$. At $(2, 3)$: $\frac{dy}{dx} = -\frac{3}{2}$.